STOP USING THE WRONG CLASSIFIER!

One Algorithm. Three Different Engines.

In Part 3, we counted simple categories like *"Sunny"* or *"Hot"*. But what if your column is **36.8°C**, or **word counts**, or **True/False flags**? You switch the Naive Bayes engine!

⚡ THE DATA-TYPE ROUTER: HOW NAIVE BAYES PICKS ITS MATH

ENGINE 01

Gaussian NB

For **continuous decimals** & measurements.

```
Temp = 36.8°C
Salary = ₹14.5L
```

ENGINE 02

Multinomial NB

For **word counts** & frequency tables.

```
"free": 5 times
"win": 3 times
```

ENGINE 03

Bernoulli NB

For strict **Yes/No (1/0)** presence flags.

```
has_link = 1
has_offer = 0
```

The Golden Rule of Naive Bayes

Bayes' Theorem (`Posterior ∝ Prior × Likelihood`) **never changes**. The only thing that changes across these 3 models is **how we calculate the Likelihood P(x | y)**!

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Swipe for Engine #1: Gaussian →

GAUSSIAN NAIVE BAYES (GAUSSIANNB)

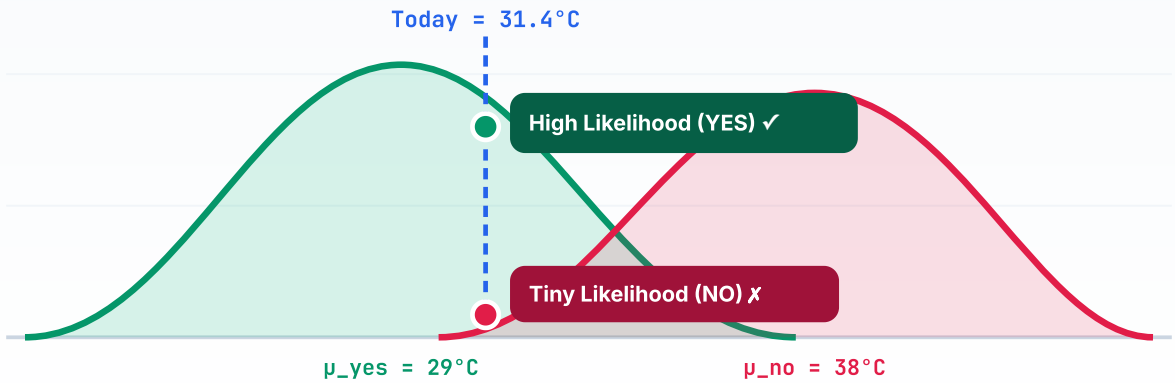
When You Can't Count Exact Matches, Use a Bell Curve!

What if today's temperature is **31.4°C**, and *no past day was ever 31.4°C*? Counting gives `0/9`! Instead, Gaussian NB fits a smooth **Bell Curve** to each class.

📈 **VISUAL INTUITION: MEASURING CURVE HEIGHT AT X = 31.4°C**

● Play Golf = YES (Mean μ = 29°C)

● Play Golf = NO (Mean μ = 38°C)



THE GAUSSIAN FORMULA (DON'T PANIC!)

$$P(x \mid y) = [1 / \sqrt{(2\pi\sigma^2)}] \cdot \exp(-(x - \mu)^2 / 2\sigma^2)$$

All this formula asks is: **"How close is today's number (x) to the class average (μ), scaled by the spread (σ)?"** Closer to the peak = higher score!

BEST REAL-WORLD USE CASES

🏥 **Medical Vitals:** Blood pressure, glucose, BMI

🏠 **Sensor & IoT Data:** Temperature, wind speed

💳 **Finance:** Transaction amounts, credit ratios

🌸 **Classic ML:** Iris petal length & width

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Swipe for Engine #2: Multinomial →

MULTINOMIAL NAIVE BAYES (MULTINOMIALNB)

The King of NLP: The More a Word Repeats, The More It Matters!

When classifying emails or reviews, seeing the word **"FREE" 5 times** is a much stronger spam clue than seeing it once. **Multinomial NB** counts total word frequencies!

📬 **INCOMING EMAIL: "WIN FREE CASH! FREE PRIZE! CLICK TO WIN!"** `CountVectorizer()`

"free"	"win"	"cash"	"meeting"
2×	2×	1×	0×
Power Exponent = 2	Power Exponent = 2	Power Exponent = 1	Ignored! (Count = 0)

HOW MULTINOMIAL NB COMPUTES WORD PROBABILITY

$$P(\text{"free"} \mid \text{Spam}) = (\text{Count of "free" in Spam} + 1) \div (\text{Total Words in All Spam} + V)$$

⚡ **SUPERPOWER: MISSING WORDS ARE IGNORED**

If your vocabulary has 50,000 words and a short tweet only uses **8 words**, Multinomial NB only multiplies probabilities for those **8 present words**! Words with count `0` are skipped.

🎯 **WHEN TO CHOOSE MULTINOMIALNB**

- 📰 **Long Articles & News Categorization**
- ⭐ **Product Review Sentiment (1–5 Stars)**
- 🔢 **CountVectorizer / TF-IDF Matrices**
- 🚫 **Rule:** Never feed negative numbers!

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Swipe for Engine #3: Bernoulli →

BERNOULLI NAIVE BAYES (BERNOULLINB)

Here, Even Missing Words Are Active Clues!

Bernoulli NB doesn't care if a word appeared 1 time or 50 times—only **Did it appear (1) or NOT (0)?** Even better: it actively scores **absent words** as evidence!

🔍 HOW BERNOULLI NB EVALUATES EVERY SINGLE FEATURE IN YOUR CHECKLIST:

`Flag = 1 (Present)` ✅

Word "urgent" IS in email

`Multiplies by: P("urgent" | y)`

Rewards classes where "urgent" commonly appears.

`Flag = 0 (Absent)` ❌

Word "unsubscribe" is MISSING

`Multiplies by: (1 - P("unsub" | y))`

Active penalty! Multinomial NB ignores 0s, but Bernoulli uses 0s as a real clue!

BERNOULLI LIKELIHOOD RULE: $P(x \mid y) = P(i \mid y)^x \cdot [1 - P(i \mid y)]^{(1 - x)}$ where $x \in \{0, 1\}$

⚡ SAME TEXT, DIFFERENT VECTORIZER INPUT:

```
MultinomialNB:
[3, 0, 5, 1, 0] # Raw counts
```

```
BernoulliNB(binarize=0.0):
[1, 0, 1, 1, 0] # Clipped to 1/0
```

BOOKMARK THIS SLIDE FOR INTERVIEWS

Which Naive Bayes Should You Pick in 10 Seconds?

Don't guess by trial and error—match the classifier directly to how your features are measured:

Feature	🔔 GaussianNB	📊 MultinomialNB	🔘 BernoulliNB
Data Type	Continuous decimals (`float`)	Discrete counts (`0, 1, 2, 5...`)	Binary flags (`0` or `1` only)
What It Learns	Mean (μ) & Std Dev (σ) per class	Word fraction of total class words	Fraction of docs containing feature
When Feature = 0	Evaluates bell curve at `x = 0`	**Ignored** (`P^0 = 1`)	**Scored!** Uses `(1 − P)`
Best For	Sensor data, medical metrics, Iris	Long articles, TF-IDF, topic modeling	Short tweets, spam flags, tabular 1/0
Key Hyperparam	`var_smoothing`	`alpha=1.0` (Laplace)	`alpha=1.0`, `binarize`

🧠 **Classic Interview Question: "What if my dataset has BOTH continuous AND binary features?"**
Because Naive Bayes assumes features are independent, you can fit **GaussianNB on the continuous columns** and **BernoulliNB on the binary columns**, then **add their `predict_log_proba()` scores together** (subtracting the duplicate class log-prior once)!

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Swipe for Python Code Template →

COPY-PASTE PYTHON TEMPLATE

PRODUCTION-READY SCIKIT-LEARN

All 3 Naive Bayes Engines in 12 Lines of Python

Notice how all three share the exact same `.fit(X, y)` and `.predict(X)` API—you just swap the class to match your feature matrix!

three_naive_bayes_flavors.py • scikit-learn

```python
from sklearn.naive_bayes import GaussianNB, MultinomialNB, BernoulliNB
from sklearn.feature_extraction.text import CountVectorizer

# 1. GAUSSIAN NB → Continuous sensor/medical measurements (floats)
gnb = GaussianNB().fit(X_continuous, y)

# 2. MULTINOMIAL NB → Word count matrix (e.g. [3, 0, 5, 1])
X_counts = CountVectorizer(binary=False).fit_transform(emails)
mnb = MultinomialNB(alpha=1.0).fit(X_counts, y_spam)

# 3. BERNOULLI NB → Binary presence/absence matrix (e.g. [1, 0, 1, 1])
X_binary = CountVectorizer(binary=True).fit_transform(emails)
bnb = BernoulliNB(alpha=1.0).fit(X_binary, y_spam)
```

DECIMALS?
Use GaussianNB

WORD COUNTS?
Use MultinomialNB

YES/NO (1/0)?
Use BernoulliNB

SERIES FINALE NEXT • PART 05

The Zero-Probability Trap (Why 1 Unseen Word Destroys Naive Bayes) & Laplace Smoothing!

Follow for Part 5 🔔

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